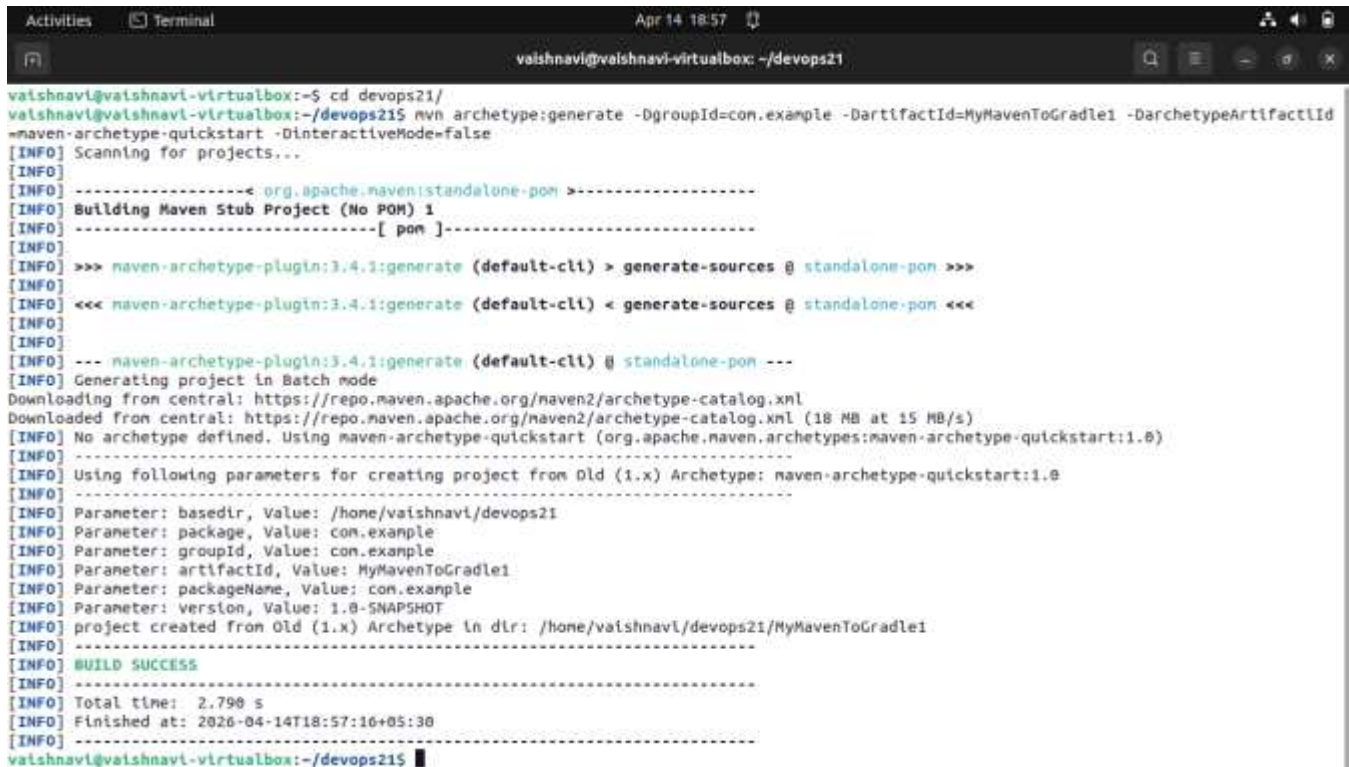


Converting Maven to Gradle

Step 1:

Create a simple maven project with the command:

```
mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenToGradle -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
```



```
vaishnavi@vaishnavi-virtualbox:~$ cd devops21/
vaishnavi@vaishnavi-virtualbox:~/devops21$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenToGradle1 -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] ----- org.apache.maven:standalone-pom -----
[INFO] Building Maven Stub Project (No POM) 1
[INFO] -----[ pom ]-----
[INFO]
[INFO] >>> mvn archetype:plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO] <<< mvn archetype:plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO] --- mvn-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml (18 MB at 15 MB/s)
[INFO] No archetype defined. Using maven-archetype-quickstart (org.apache.maven.archetypes:maven-archetype-quickstart:1.0)
[INFO]
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO]
[INFO] Parameter: basedir, Value: /home/vaishnavi/devops21
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenToGradle1
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/vaishnavi/devops21/MyMavenToGradle1
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 2.790 s
[INFO] Finished at: 2026-04-14T18:57:16+05:30
[INFO]
vaishnavi@vaishnavi-virtualbox:~/devops21$
```

The command can be broken down as:

- DgroupId=com.example – project’s base package (reverse-domain style).
- DartifactId=my-maven-app – defines the name of the project (will also be the folder name).
- DarchetypeArtifactId=maven-archetype-quickstart – sets the Maven template to be the basic Java application.
- DinteractiveMode=false – deactivates interactive mode and uses the given values automatically.

Step 2:

Edit the file pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-v4_0_0.xsd">
  <modelVersion>4.0.0</modelVersion>
  <groupId>com.example</groupId>
  <artifactId>MyMavenApp</artifactId>
  <packaging>jar</packaging>
```

```
<version>1.0-SNAPSHOT</version>
```

```
<name>MyMavenApp</name>
```

```
<url>http://maven.apache.org</url>
```

```
<!-- Properties: Customize Java version or plugin versions -->
```

```
<properties>
```

```
<maven.compiler.source>11</maven.compiler.source>
```

```
<maven.compiler.target>11</maven.compiler.target>
```

```
</properties>
```

```
<dependencies>
```

```
<dependency>
```

```
<groupId>junit</groupId>
```

```
<artifactId>junit</artifactId>
```

```
<version>3.8.1</version>
```

```
<scope>test</scope>
```

```
</dependency>
```

```
</dependencies>
```

```
<!-- Build: Configuring plugins and build settings -->
```

```
<build>
```

```
<plugins>
```

```
<!-- Example: Maven Compiler Plugin to compile Java code -->
```

```
<plugin>
```

```
<groupId>org.apache.maven.plugins</groupId>
```

```
<artifactId>maven-compiler-plugin</artifactId>
```

```
<version>3.8.1</version>
```

```
<configuration>
```

```
<source>1.7</source>
```

```
<target>1.7</target>
```

```
</configuration>
```

```
</plugin>
```

<!-- Example: Maven Surefire Plugin to run tests -->

<plugin>

<groupId>org.apache.maven.plugins</groupId>

<artifactId>maven-surefire-plugin</artifactId>

<version>2.22.2</version>

</plugin>

<plugin>

<groupId>org.apache.maven.plugins</groupId>

<artifactId>maven-jar-plugin</artifactId>

<version>3.1.0</version>

<configuration>

<archive>

<manifestEntries>

<Main-Class>com.example.MainClass</Main-Class>

</manifestEntries>

</archive>

</configuration>

</plugin>

</plugins>

</build>

</project>

Step 3:

Build the project using the command :

\$mvn clean install

```

Activities Terminal Apr 14 19:20
vaishnavi@vaishnavi-virtualbox: ~/devops21/MyMavenToGradle1

[INFO] --- maven-clean-plugin:2.5:clean (default-clean) @ MyMavenApp ---
[INFO] Deleting /home/vaishnavi/devops21/MyMavenToGradle1/target
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MyMavenApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/vaishnavi/devops21/MyMavenToGradle1/src/main/resources
[INFO] --- maven-compiler-plugin:3.8.1:compile (default-compile) @ MyMavenApp ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to /home/vaishnavi/devops21/MyMavenToGradle1/target/classes
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ MyMavenApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/vaishnavi/devops21/MyMavenToGradle1/src/test/resources
[INFO] --- maven-compiler-plugin:3.8.1:testCompile (default-testCompile) @ MyMavenApp ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to /home/vaishnavi/devops21/MyMavenToGradle1/target/test-classes
[INFO] --- maven-surefire-plugin:2.22.2:test (default-test) @ MyMavenApp ---
[INFO]
[INFO]
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.003 s - in com.example.AppTest
[INFO] Results:
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] --- maven-jar-plugin:3.1.0:jar (default-jar) @ MyMavenApp ---
[INFO] Building jar: /home/vaishnavi/devops21/MyMavenToGradle1/target/MyMavenApp-1.0-SNAPSHOT.jar
[INFO] --- maven-install-plugin:2.4:install (default-install) @ MyMavenApp ---
[INFO] Installing /home/vaishnavi/devops21/MyMavenToGradle1/target/MyMavenApp-1.0-SNAPSHOT.jar to /home/vaishnavi/.m2/repository/com/example/MyMavenApp/1.0-SNAPSHOT/MyMavenApp-1.0-SNAPSHOT.jar
[INFO] Installing /home/vaishnavi/devops21/MyMavenToGradle1/pom.xml to /home/vaishnavi/.m2/repository/com/example/MyMavenApp/1.0-SNAPSHOT/MyMavenApp-1.0-SNAPSHOT.pom
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 2.374 s
[INFO] Finished at: 2025-04-14T19:20:05+05:30
[INFO]
vaishnavi@vaishnavi-virtualbox: ~/devops21/MyMavenToGradle1$

```

If the command shows output as “build success”, it means that the project has compiled successfully

Step 4:

Migrating the maven project to gradle.

Execute the commands given below:

\$git init --type pom.xml

```

[INFO] -----
vaishnavi@vaishnavi-virtualbox: ~/devops21/MyMavenToGradle1$ gradle init --type pom.xml
Starting a Gradle Daemon, 1 incompatible and 1 stopped Daemons could not be reused, use --status for details

FAILURE: Build failed with an exception.

* What went wrong:
Execution failed for task ':init'.
> The requested build setup type 'pom.xml' is not supported. Supported types: 'basic', 'groovy-application', 'groovy-library', 'java-application', 'java-library', 'pom', 'scala-library'.

* Try:
Run with --stacktrace option to get the stack trace. Run with --info or --debug option to get more log output. Run with --scan to get full insights.

* Get more help at https://help.gradle.org

BUILD FAILED in 3s
2 actionable tasks: 2 executed
vaishnavi@vaishnavi-virtualbox: ~/devops21/MyMavenToGradle1$

```

Error explanation:

- ❑ pom.xml is the name of a configuration file used in Maven projects, not a project type. Gradle does not recognize file names in the --type option.
- ❑ The --type flag in Gradle expects a valid project type keyword (like java-application, basic, or pom) that tells Gradle what kind of project to create.
- ❑ Since pom.xml is not in Gradle’s list of supported types, it throws an error saying it is not supported.

- ❓ On the other hand, pom is a valid type in Gradle, used when you want to create a project compatible with Maven-style structure.

Step 5:

Execute the command to view the files

\$ls

There is no build.gradle file in the project.

Step 6:

To correct the error, execute the command:

\$git init --type pom

```
2 actionable tasks: 2 executed
vaishnavi@vaishnavi-virtualbox:~/devops21/MyMavenToGradle1$ ls
gradle gradlew gradlew.bat pom.xml src target
vaishnavi@vaishnavi-virtualbox:~/devops21/MyMavenToGradle1$ gradle init --type pom

> Task :init
Maven to Gradle conversion is an incubating feature.

BUILD SUCCESSFUL in 2s
2 actionable tasks: 1 executed, 1 up-to-date
vaishnavi@vaishnavi-virtualbox:~/devops21/MyMavenToGradle1$
```

gradle init --type pom is used to initialize (create) a new Gradle project

It tells Gradle to create the project in a Maven-style structure (similar to how Maven projects are organized)

It generates basic files like:

- build.gradle (Gradle build file)
- project folder structure (like src/main/java, etc.)

The pom type specifically is used when you want a simple project setup or a parent project (like Maven's pom packaging)

Step 6:

Execute the same command again or the given command to view the project structure.

\$tree

```

vaishnavi@vaishnavi-virtualbox: ~/devops21/MyMavenToGradle1$ tree
.
├── build.gradle
├── gradle
│   └── wrapper
│       ├── gradle-wrapper.jar
│       └── gradle-wrapper.properties
├── gradlew
├── gradlew.bat
├── pom.xml
├── settings.gradle
├── src
│   ├── main
│   │   ├── java
│   │   │   ├── com
│   │   │   │   └── example
│   │   │       └── App.java
│   └── test
│       ├── java
│       │   ├── com
│       │   │   └── example
│       │       └── AppTest.java
├── target
│   ├── classes
│   │   ├── com
│   │   │   └── example
│   │       └── App.class
│   ├── generated-sources
│   │   └── annotations
│   ├── generated-test-sources
│   │   └── test-annotations
│   ├── maven-archiver
│   │   └── pom.properties
│   ├── maven-status
│   │   └── maven-compiler-plugin
│   │       ├── compile
│   │       │   ├── default-compile
│   │       │   │   ├── createdFiles.lst
│   │       │   │   └── inputFiles.lst
│   │       └── testCompile
│   │           ├── default-testCompile
│   │           │   ├── createdFiles.lst
│   │           │   └── inputFiles.lst
│   └── MyMavenApp-1.0-SNAPSHOT.jar
├── surefire-reports
│   ├── com.example.AppTest.txt
│   └── TEST-com.example.AppTest.xml
└── test-classes

```

Now, we can make the given observation in the output:

This time build.gradle file is created n included in the file

Step 7:

Push the project into the github using the commands:

```
$ git config --global user.name "vaishnavim1121"
```

```
$ git config --global user.email "vaishnavim1121@gmail.com"
```

```
$ git init $ git add . $ git commit -m "initial commit"
```

```
$ git remote add origin https://github.com/vaishnavim1121/MyMavenToGradle.git
```

```
$ git remote set-url origin https://\*PAT\*@github.com/vaishnavim1121/MyMavenToGradle
```

```
$ git push -u origin master
```

```

Activities Terminal Apr 14 19:10
vaishnavi@vaishnavi-virtualbox: ~/devops21/MyMavenToGradle1
AppTest.class

38 directories, 18 files
vaishnavi@vaishnavi-virtualbox:~/devops21/MyMavenToGradle1$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/vaishnavi/devops21/MyMavenToGradle1/.git/
vaishnavi@vaishnavi-virtualbox:~/devops21/MyMavenToGradle1$ git add .
vaishnavi@vaishnavi-virtualbox:~/devops21/MyMavenToGradle1$ git commit -m 'maven to gradle'
[master (root-commit) 255ca2d] maven to gradle
27 files changed, 473 insertions(+)
create mode 100644 .gradle/4.4.1/fileChanges/last-build.bin
create mode 100644 .gradle/4.4.1/fileHashes/fileHashes.bin
create mode 100644 .gradle/4.4.1/fileHashes/fileHashes.lock
create mode 100644 .gradle/4.4.1/taskHistory/taskHistory.bin
create mode 100644 .gradle/4.4.1/taskHistory/taskHistory.lock
create mode 100644 .gradle/buildOutputCleanup/buildOutputCleanup.lock
create mode 100644 .gradle/buildOutputCleanup/cache.properties
create mode 100644 .gradle/buildOutputCleanup/outputFiles.bin
create mode 100644 build.gradle
create mode 100644 gradle/wrapper/gradle-wrapper.jar
create mode 100644 gradle/wrapper/gradle-wrapper.properties
create mode 100755 gradlew
create mode 100644 gradlew.bat
create mode 100644 pom.xml
create mode 100644 settings.gradle
create mode 100644 src/main/java/com/example/App.java
create mode 100644 src/test/java/com/example/AppTest.java
create mode 100644 target/myMavenApp-1.0-SNAPSHOT.jar
create mode 100644 target/classes/com/example/App.class
create mode 100644 target/maven-archiver/pom.properties
create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/createdFiles.lst
create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/inputFiles.lst
create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/createdFiles.lst
create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/inputFiles.lst
create mode 100644 target/surefire-reports/TEST-com.example.AppTest.xml
create mode 100644 target/surefire-reports/com.example.AppTest.txt
create mode 100644 target/test-classes/com/example/AppTest.class
vaishnavi@vaishnavi-virtualbox:~/devops21/MyMavenToGradle1$ git remote add origin https://github.com/vaishnavi121/MyMavenToGradle1.git
git: 'remote' is not a git command. See 'git --help'.

remote
vaishnavi@vaishnavi-virtualbox:~/devops21/MyMavenToGradle1$ git remote add origin https://github.com/vaishnavi121/MyMavenToGradle1.git
vaishnavi@vaishnavi-virtualbox:~/devops21/MyMavenToGradle1$ git remote set-url origin https://ghp_NMyPMp5Sxgt2QwFOV7noGjtJaL6ZV3wWfI@github.com/vaishnavi121/MyMavenToGradle1
vaishnavi@vaishnavi-virtualbox:~/devops21/MyMavenToGradle1$ git push -u origin master
vaishnavi@vaishnavi-virtualbox:~/devops21/MyMavenToGradle1$ git push -f origin master
Enumerating objects: 61, done.
Counting objects: 100% (61/61), done.
Delta compression using up to 10 threads
Compressing objects: 100% (34/34), done.
Writing objects: 100% (61/61), 61.01 KiB | 8.72 MiB/s, done.
Total 61 (delta 2), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (2/2), done.
To https://github.com/vaishnavi121/MyMavenToGradle1
+ 17f5a91...255ca2d master -> master (forced update)
vaishnavi@vaishnavi-virtualbox:~/devops21/MyMavenToGradle1$
vaishnavi@vaishnavi-virtualbox:~/devops21/MyMavenToGradle1$

```